ANSH SHAH

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EDUCATION

Brock University, St. Catharines, ON Sept 2022 - June 2026
BSc (Honours) Computer Science, Concentration in Software Engineering
Undergraduate Thesis: conformal prediction for reliable bug triage

PUBLICATIONS

MACCP: Agreement-Conditioned Conformal Prediction for Bug Triage
Ansh Shah *(first author)*, Naser Ezzati-Jivan | 26th IEEE Int'l Conf. on Software Quality, Reliability and Security (QRS 2026), Springer LNCS | Code — Florence, July 2026

Sequence-Based Evolutionary and Neural Strategies for Reducing Zone Crossings in Toolpaths
Y. Aali, **A. Shah**, M.I. Uddin, K.N. Anwar, S. Houghten, R.I. Nishat | 39th Canadian Conf. on Artificial Intelligence (Canadian AI 2026), PMLR vol. 318 | Code — Vancouver, May 2026

EXPERIENCE

Brock University, *Undergraduate Researcher, Machine Learning for Software Engineering* Sept 2025 - June 2026

- First-authored a conformal prediction framework for bug triage (IEEE QRS 2026): fine-tuned DeBERTa-v3 transformers and engineered XGBoost pipelines over 226K+ bug reports across three industrial datasets, producing calibrated prediction sets that auto-triage 29% of incoming bugs at 90% empirical accuracy under 15+ years of temporal distribution shift.
- Ran all experiments on multi-GPU HPC clusters (4x NVIDIA H100, SLURM): distributed transformer fine-tuning, 100K-sample dataset generation pipelines, DeepSpeed ZeRO LoRA fine-tuning of 32B-parameter LLMs, and ablation suites from zero-shot LLMs to gradient-boosted baselines.

Brock University, *Research Collaborator, Neural Optimization for 3D Printing* Jan 2026 - May 2026

- Designed and built FusionNet, the complete neural component of a Canadian AI 2026 paper: a hybrid CNN-RNN (Residual U-Net, GRU, FiLM conditioning, multi-head self-attention) that predicts toolpath reconfiguration operations, trained on 22K+ optimization trajectories.
- Iterated four architecture generations and an alternating model-SA inference loop, cutting optimizer operation counts by up to 96% and runtime by 6.7x against simulated-annealing baselines.

Wolf-Rayet Technology, *Full-stack Developer* | India | Design Demo Dec 2021 - May 2022

- Delivered full-cycle web projects for multiple clients, including a US$5,000 custom contract taken from design through deployment.
- Hardened production backends with server-side input validation and sanitization against XSS and SQL injection, and built secure authentication with session management and Twilio-backed password recovery.
- Designed the company's CI/CD pipeline with Cron-scheduled automated backups for error-free client deployments.

PROJECTS

Multi-Agent AI Coding Assistant (opencode fork) | *TypeScript, Bun, SolidJS, SQLite* | Github 2026 - Present

- Extended opencode, an open-source terminal-first AI coding agent, into a multi-agent system: an orchestration engine that runs specialist sub-agents (reviewer, researcher, tester, refactor) under parallel, pipeline, map-reduce, and consensus strategies.
- Built context intelligence for long coding sessions: automatic project-context injection (git state, framework detection), persistent project memory, and dynamic context pruning that fits large histories into model context windows.
- Added structured git/test/lint tools, an interactive PTY, automatic secret redaction on tool output, and a split-pane TUI cockpit with live sub-agent monitoring, diff viewer, and interactive plan review.

Academic FBI, Plagiarism Detection Platform | *TypeScript, React, Python, Docker* | Live | Github Sept 2025 - April 2026

- Built a plagiarism-detection platform for academic code submissions with a 6-person team: instructors upload class submissions, run similarity analysis, and review flagged pairs in a side-by-side diff viewer.
- Collaborated through a shared pull-request workflow across two terms: authored 9 of the team's 18 PRs, integrated and merged teammates' branches, and set up the GitHub Actions CI pipeline the whole team shipped through.
- Owned the React/TypeScript frontend and integrated it against the team's Python analysis backend, contributing backend fixes and tests across the shared codebase.

SKILLS

Languages: Python, Java, C/C++, TypeScript, JavaScript, SQL, PHP, Bash, HTML/CSS, Assembly (x86)
ML & Data: PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, XGBoost, pandas, NumPy, DeepSpeed, LoRA
Software Tools: Git, Linux, Docker, SLURM (HPC), CI/CD, CMake, Bazel, LaTeX, Insomnia